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In [ ]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

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In [ ]: df = sns.load_dataset('titanic')
```

```
In [ ]: print("Sample Data:")
print(df.head())
```

Sample Data:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

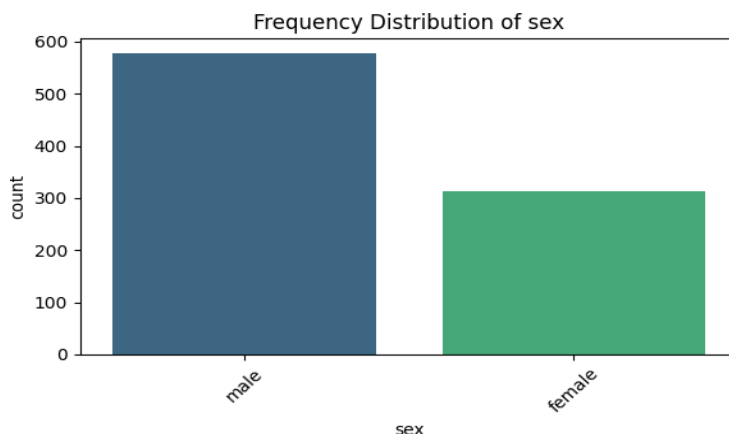
	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

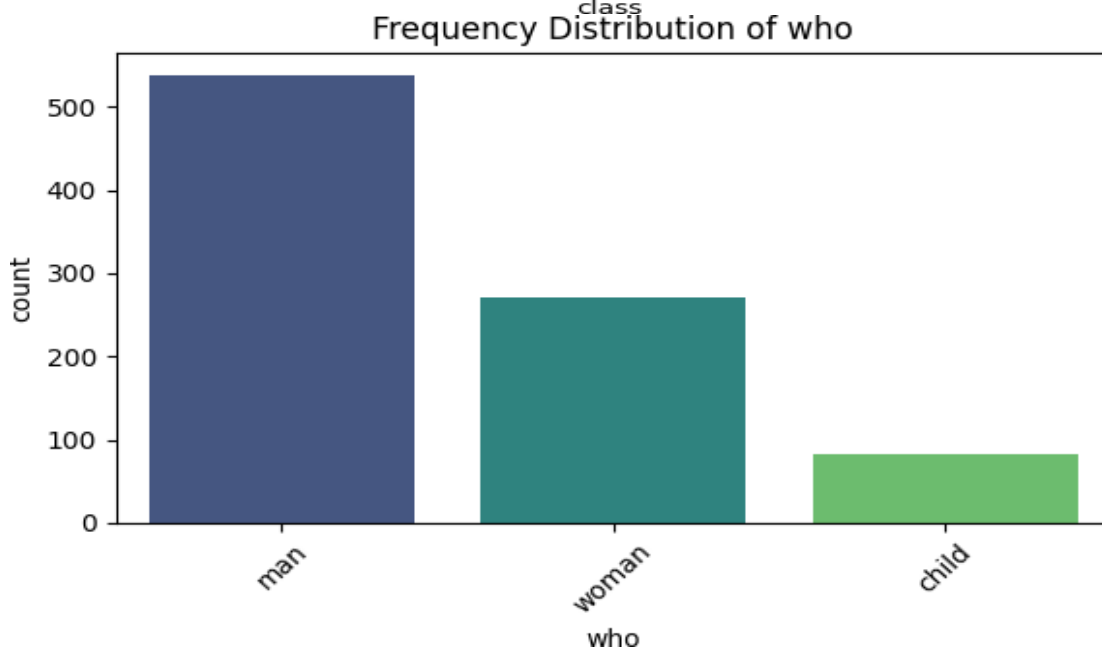
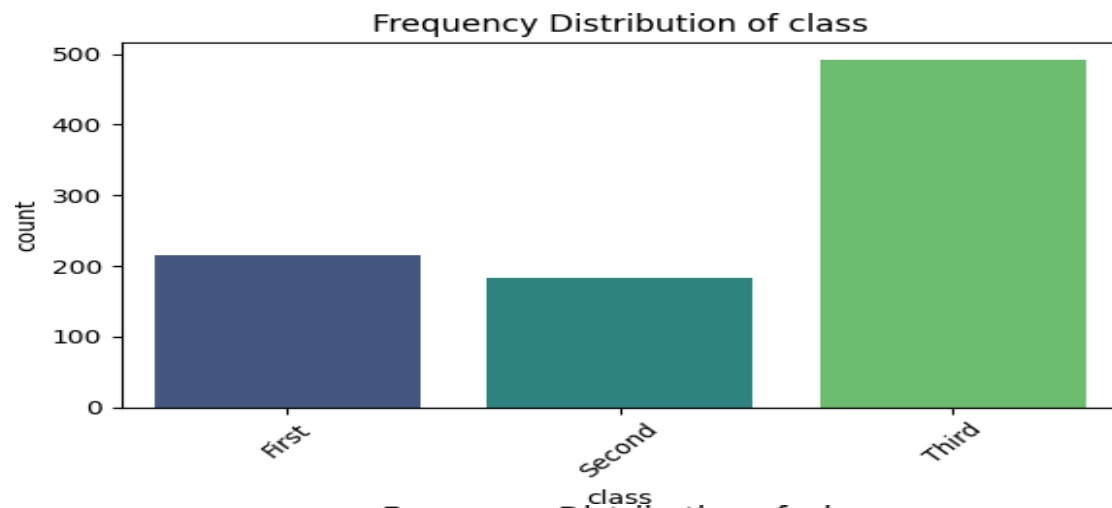
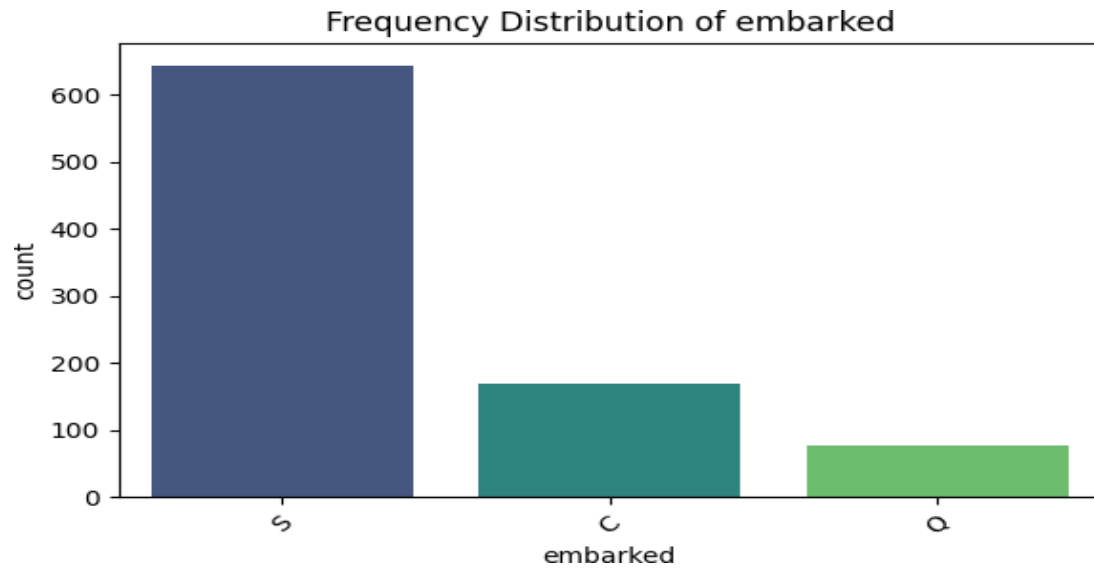
```
In [ ]: categorical_cols = df.select_dtypes(include=['object', 'category']).columns
print("\nCategorical Columns:")
print(categorical_cols)
```

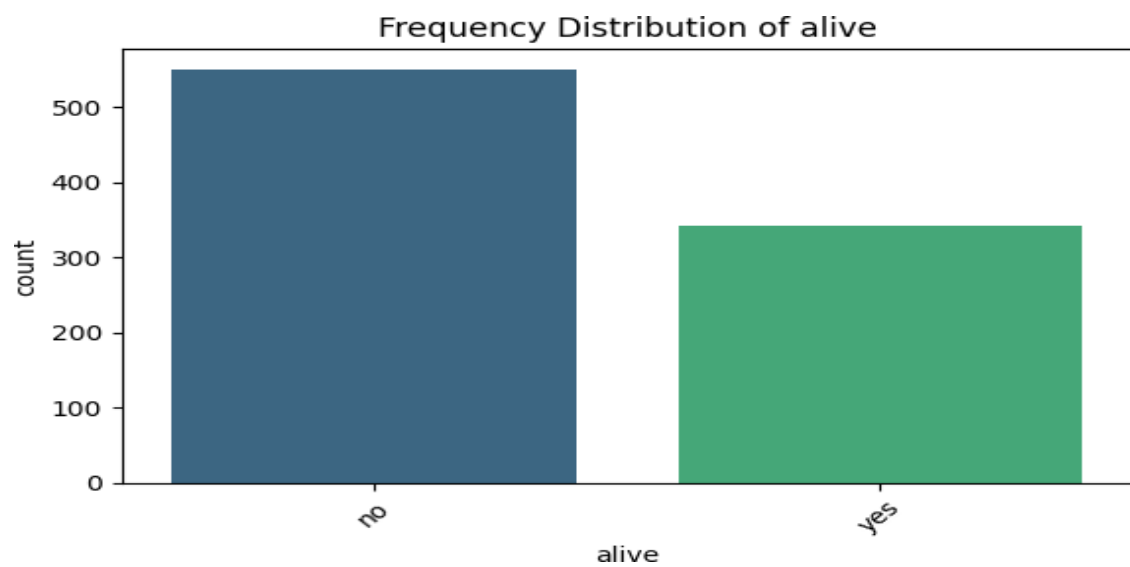
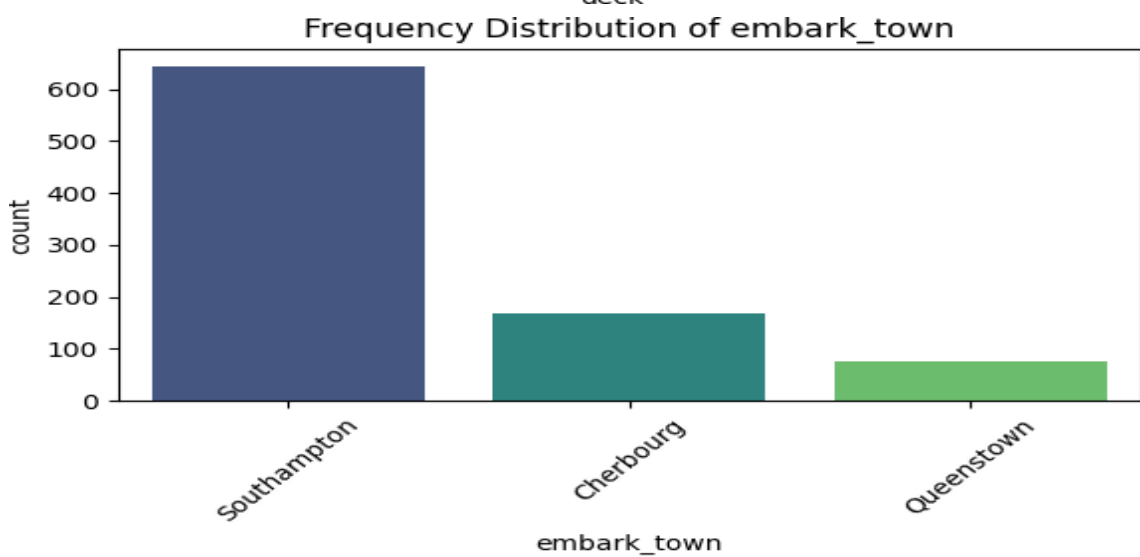
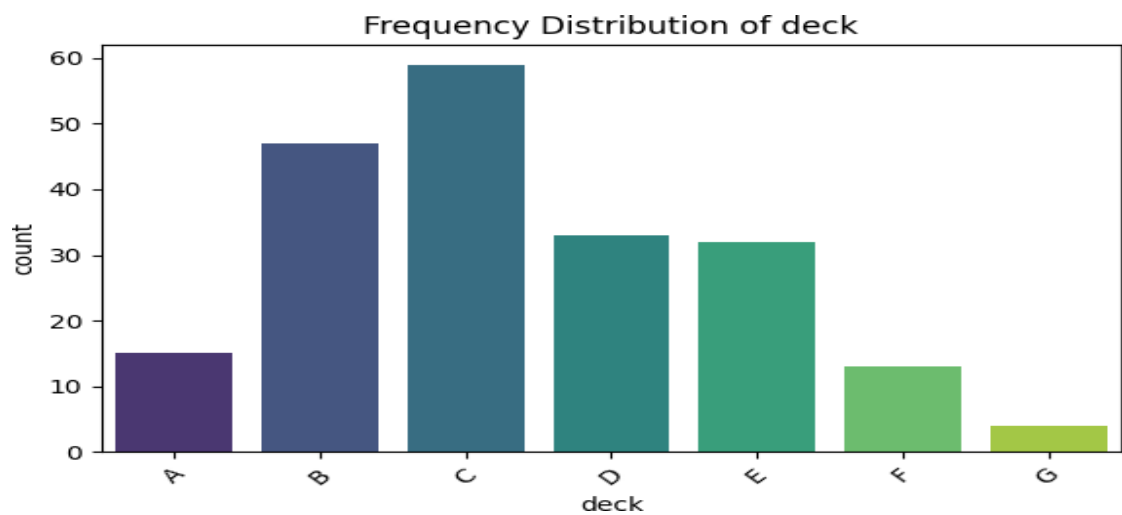
Categorical Columns:

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Index(['sex', 'embarked', 'class', 'who', 'deck', 'embark_town', 'alive'], dtype='object')
```

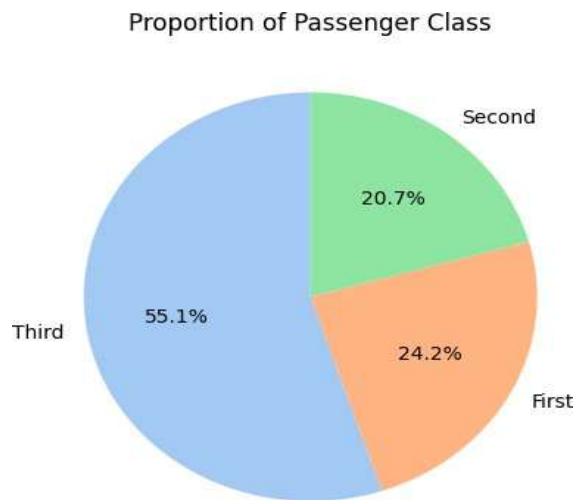
```
In [ ]: for col in categorical_cols:
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x=col, palette='viridis', legend=False)
plt.title(f"Frequency Distribution of {col}")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```







```
In [ ]: plt.figure(figsize=(5, 5))
df['class'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90,
                                     colors=sns.color_palette('pastel'))
plt.title("Proportion of Passenger Class")
plt.ylabel('')
plt.show()
```



```
In [ ]: plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='sex', hue='class', palette='Set2')
plt.title("Distribution of Passenger Class Split by Sex")
plt.xlabel("Sex")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```

